WQD7001 - PRINCIPLES OF DATA SCIENCE

GROUP ASSIGNMENT 2

Predictive Analytics of Diabetes

Using Health Survey Data

GROUP 11

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CHAPTER 1 : INTRODUCTION

1.1 Background

The Diabetes Health Indicators dataset is a cleaned and structured version of the 2015 BRFSS (Behavioral Risk Factor Surveillance System) survey conducted by the CDC in the United States. This dataset, originally sourced from Kaggle and UCI Machine Learning Repository, contains 253,680 respondent records and 21 health-related features. These features include demographic factors, lifestyle behaviors, and chronic health conditions such as high blood pressure, high cholesterol, smoking habits, physical activity, and BMI. The dataset provides a rich foundation for analysing how non-clinical, survey-based indicators can be used to assess and predict diabetes risk.

Understanding diabetes risk using survey data is extremely valuable because such data does not require lab tests or clinical devices. Instead, it relies on self-reported behaviors and general health indicators, making it easier and cheaper to collect on a large scale. This makes predictive modeling based on this dataset a potential low-cost, scalable solution for early diabetes detection and preventive health programs. Identifying patterns in lifestyle and demographic factors can help public health agencies improve awareness and intervention strategies.

1.2 Objectives

1.2.1 Primary Objective

1. To develop and evaluate a robust machine-learning framework for the early prediction of diabetes, leveraging survey-based health indicators to identify high-risk individuals and provide interpretable diagnostic insights.

1.2.2 Specific Objectives

1. Data Curative and Preprocessing: To implement rigorous data cleaning and preprocessing pipelines, addressing missing values and outliers in health survey data.

2. Exploratory and Correlation Analysis: To perform comprehensive Exploratory Data Analysis (EDA) to identify significant demographic and physiological correlations associated with diabetic risk.

3. Comparative Model Development: To design and train a suite of multiclass classification algorithms (Random Forest, XGBoost, Logistic Regression) to identify predictive performance.

4. Performance Validation: To quantitatively assess model reliability using a multi-metric approach, including F1-score, ROC-AUC, and precision-recall trade-offs.

5. Interpretability and Risk Factor Analysis: To extract and analyze feature importance to determine the primary clinical drivers of diabetes risk.

CHAPTER 2 : EXPLORATORY DATA ANALYSIS

2.1 Target Variable Analysis

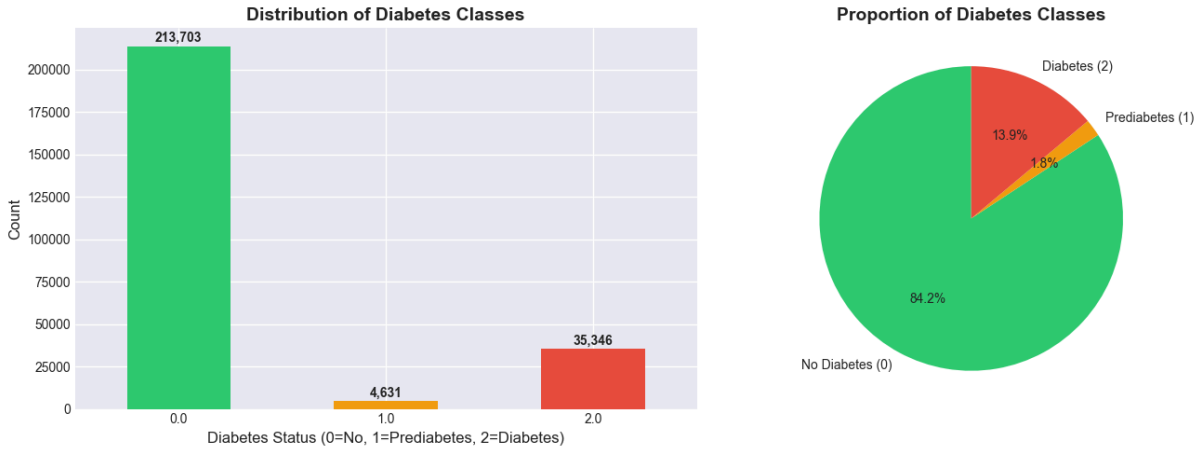

Figure 1 : *Target Variable Analysis Bar Chart & Pie Chart*

The target variable in the dataset shows a notable class imbalance. Non-diabetic people (Status=0) make up the great majority of the cohort, accounting for 84.2% of the total (n=213,703). On the other hand, the prediabetic minority (Status=1) makes up only 1.8% (n=4,631), but the diabetic group (Status=2) constitutes 13.9% (n=35,346). To ensure that the

minority classes are correctly predicted, this distribution requires the application of particular modeling techniques.

2.2 Numerical Features Analysis

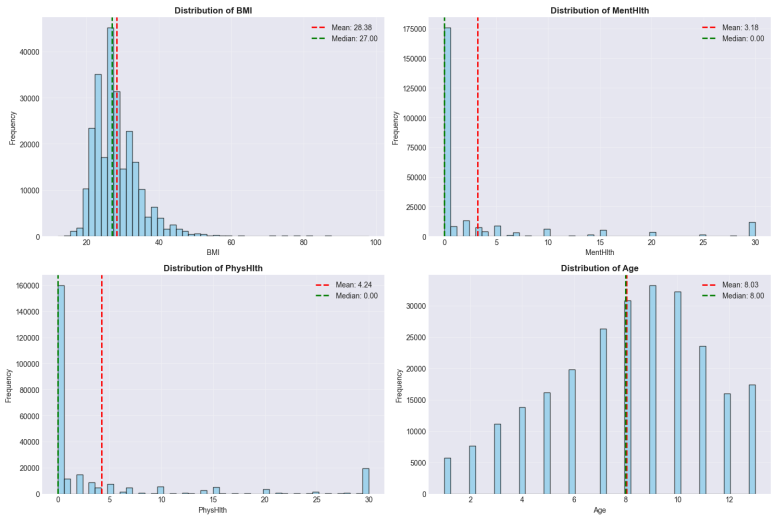

Figure 2 : *Distributions of different discrete variables (age, physical health, and mental health)*

Three discrete variables (age, physical health, and mental health) and one continuous variable (BMI) make up the numerical features. With a mean of 28.38 and a median of 27, the BMI distribution analysis indicates that about half of the participants are overweight or obese. The majority of the cohort appears to reflect high levels of health, however, as evidenced by the Mental Health and Physical Health indicators, which are significantly biased toward the lower end of the scale. In terms of the age distribution, 50% of the sample population is 55 years of age or older, according to the median value of 8 on the AGEG5YR scale.

2.3 Categorical Features Analysis

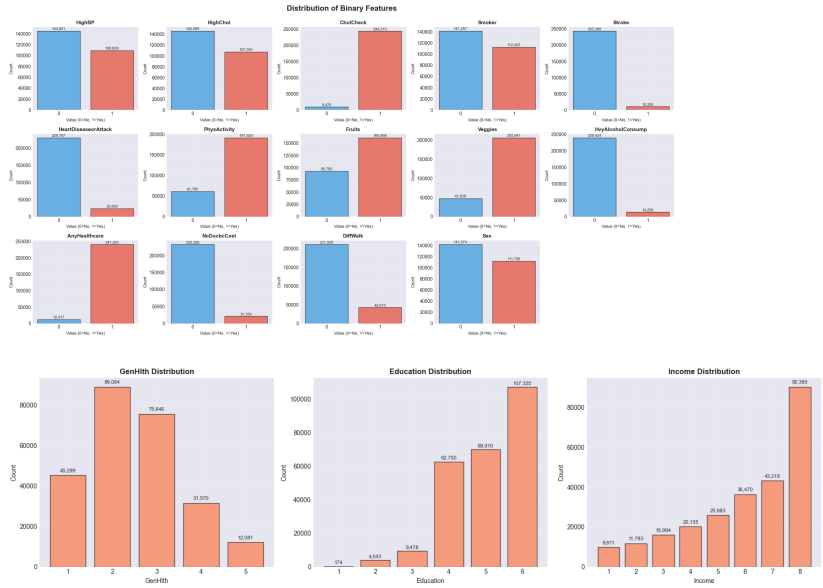

Figure 3 : *Analysis of all 14 binary features*

There are fourteen binary features in the dataset, with values of 0 and 1 denoting the existence or absence of a certain trait respectively. The majority of these characteristics are strongly dominated by the "presence" class, which indicates a general lack of balance (1). The significant exception is the Smoker variable, which shows a distribution between the two classes that is comparatively balanced.

The ordinal variables, General Health, Education, and Income reveals a cohort that is relatively affluent and well-educated. General Health (GenHlth) ratings indicate that a significant majority of participants perceive themselves to be in good physical condition. This is supported by the Education and Income metrics where the most frequent responses are concentrated in the highest categories (Level 6 for Education and Level 8 for Income). Collectively, these trends suggest that the dataset is skewed toward a higher socio-economic demographic.

2.4 Bivariate Features Analysis

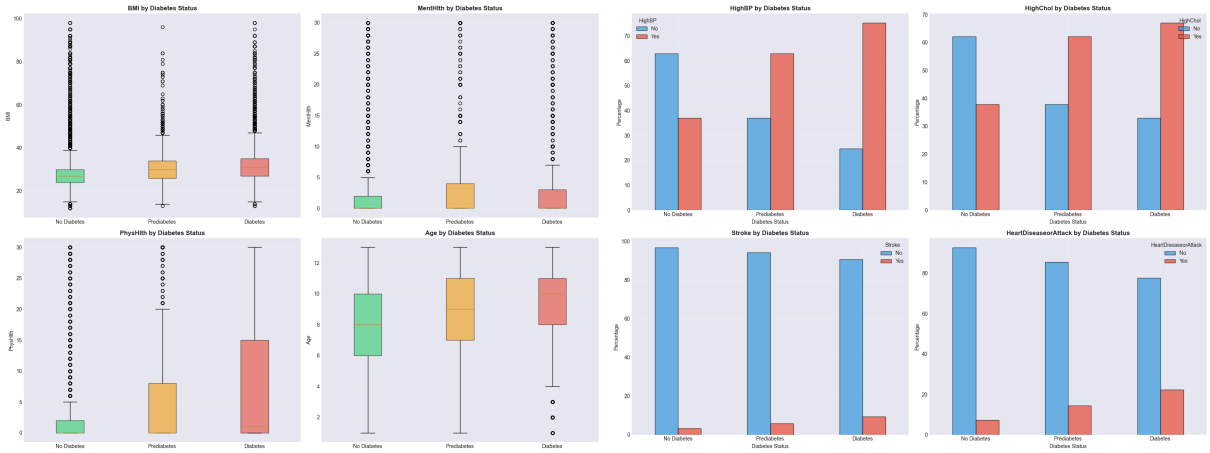

Figure 4 : *Analysis of all 14 binary features*

BMI, PhysHlth, and age are important markers of diabetic status, according to bivariate analysis; greater values are clearly associated with a higher prevalence of the condition. MentHlth, on the other hand, exhibits no variation between the "No Diabetes," "Prediabetes," and "Diabetes" groups, suggesting that there is little differentiation for this particular trait across the target labels.

Diabetes is consistently positively correlated with categorical health indicators, such as high blood pressure (HighBP), high cholesterol (HighChol), stroke, and heart disease. The diabetic group had a disproportionately greater prevalence of all four characteristics (Class 1), highlighting their importance as risk factors. The prevalence of these disorders varies, though; heart disease and stroke are much less common than high blood pressure and high cholesterol in the cohort as a whole. This implies that cardiovascular events, such as heart disease and stroke, are more specialized, lower-frequency problems in the dataset, but hypertension and dyslipidemia may function as key indicators.

CHAPTER 3 : DATA MODELING

3.1 Modelling Objective

The objective of the modelling phase is to develop a robust and generalizable machine learning model to classify individuals according to their diabetes risk based on behavioral, demographic, and health-related indicators from the BRFSS 2015 dataset. The model is intended to support population-level risk screening and early intervention strategies, rather than to provide clinical diagnosis. Specifically, the modelling phase aims to:

- Train and compare multiple classification algorithms,
- Evaluate model performance using appropriate validation techniques,
- Select the model that provides the best trade-off between predictive performance and generalization,
- Analyse feature importance to understand the key drivers of diabetes risk.

3.2 Problem Formulation

This project is formulated as a supervised classification problem, where the target variable represents diabetes status. The input features consist of health behaviors, clinical measurements, and demographic attributes.

- Target variable: Diabetes classification label
- Feature type: Numerical and encoded categorical variables
- Learning type: Supervised learning

3.3 Data Preparation for Modelling

The dataset was split into training, validation, and test sets to ensure unbiased evaluation. Prior to modelling:

- Data cleaning to remove missing values, duplicated values and carry out data type validation.
- Feature engineering to add predictive features to refine model learning.
- Feature selection using ANOVA F-test to select top 20 features.
- Features were scaled using StandardScaler to normalize numerical ranges.
- Categorical variables were encoded during preprocessing.

3.4 Model Selection

Three classification algorithms were selected:

MODEL	JUSTIFICATION
Logistic Regression	Provides a simple, interpretable baseline
Random Forest	Captures non-linear relationships and feature interactions
XGBoost	Advanced gradient boosting with regularisation to prevent overfitting

Table 1 : *Model used and their justifications*

These models allow comparison between linear and non-linear approaches and between simple and ensemble learners.

3.5 Model Training and Validation

The training dataset was used to fit each model, while the validation dataset was used for model selection. Cross-validation was applied to reduce overfitting and ensure robustness.

Performance was primarily evaluated using ROC-AUC and accuracy, as they provide a balanced assessment of classification capability, especially for imbalanced health datasets.

3.6 Model Evaluation

Each model was evaluated on the validation set using Accuracy, ROC-AUC, Score and Confusion Matrix.

The ROC-AUC metric was used as the primary selection criterion because it measures the model's ability to distinguish between classes independent of classification thresholds.

3.7 Final Model Selection

The model with the highest ROC-AUC score on the validation set was selected as the final model. In this case, the XGBoost model achieved superior predictive performance and generalization capability. Therefore, this model was selected for final testing and deployment.

3.8 Model Performance

```
============================================================
Classification Report - Logistic Regression
============================================================
              precision    recall  f1-score   support

 No Diabetes       0.86      0.97      0.92     32061
 Prediabetes       0.00      0.00      0.00       695
    Diabetes       0.53      0.18      0.27      5303

    accuracy                           0.85     38059
   macro avg       0.46      0.39      0.40     38059
weighted avg       0.80      0.85      0.81     38059

============================================================
Classification Report - Random Forest
============================================================
              precision    recall  f1-score   support

 No Diabetes       0.86      0.98      0.92     32061
 Prediabetes       0.00      0.00      0.00       695
    Diabetes       0.57      0.14      0.22      5303

    accuracy                           0.85     38059
   macro avg       0.48      0.37      0.38     38059
weighted avg       0.80      0.85      0.80     38059

============================================================
Classification Report - XGBoost
============================================================
              precision    recall  f1-score   support

 No Diabetes       0.86      0.98      0.92     32061
 Prediabetes       0.00      0.00      0.00       695
    Diabetes       0.57      0.17      0.26      5303

    accuracy                           0.85     38059
   macro avg       0.48      0.38      0.39     38059
weighted avg       0.81      0.85      0.81     38059
```

Figure 5 : *Classification Report*

OPTIMAL MODEL	RATIONALE

XGBoost	Achieved the highest Macro-F1 score (0.48), indicating the best balanced performance across all classes including minority classes.
Logistic Regression	Comparable overall accuracy (0.80) with high interpretability and transparency, making it suitable for explainable health analytics.
Random Forest	Demonstrated stable and robust performance with strong majority-class accuracy and resistance to noise.

Table 2 : *Model Performance Comparison*

CHAPTER 4 : DATA INTERPRETATION

4.0 Model Results

This study uses survey-based lifestyle, health, and demographic data from the BRFSS 2015 dataset (253,680 records with 29 engineered features) to predict diabetes risk. The objective of the model is not to replace medical diagnosis, but to act as a population-level risk screening and decision-support tool that helps identify high-risk groups and understand the key factors associated with diabetes.

The modelling task was framed as a multiclass classification problem, where individuals with prediabetes or diabetes were grouped as the positive class, and non-diabetic individuals as the negative class. Approximately 29.7% of the sample belongs to the positive class, indicating a moderately imbalanced dataset .

As displayed in Figure 6, three models were evaluated which are Logistic Regression, Random Forest, and XGBoost. Among them, XGBoost achieved the best overall validation performance with an accuracy of approximately 0.85 and a ROC-AUC of 0.819, indicating good overall discriminative ability.

```
Model Performance Comparison:
               Model  Accuracy  ROC-AUC
Logistic Regression  0.846738 0.813792
      Random Forest  0.848262 0.817107
            XGBoost  0.849760 0.819236
```

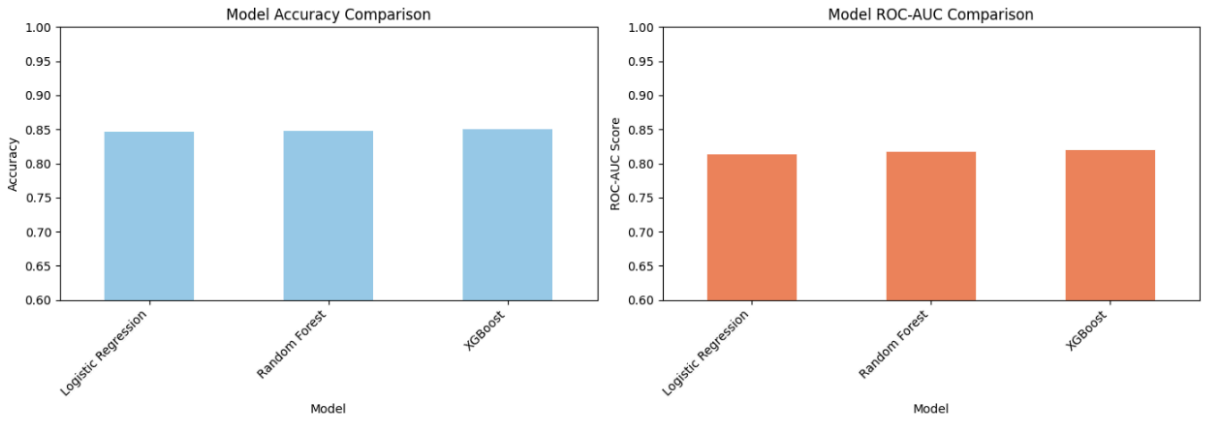

Figure 6 : *Model Performance Comparison*

However, detailed class-wise evaluation shows that while the model performs very well in identifying non-diabetic individuals (recall ≈ 0.98), it performs poorly on the prediabetic subgroup (recall ≈ 0) and only moderately on the diabetic group (recall ≈ 0.17). This means the model is more suitable for identifying general risk patterns and supporting population-level screening, rather than for early individual-level diagnosis.

4.1 Key Predictors

Feature importance analysis across Random Forest, XGBoost, and Logistic Regression shows that the most influential predictors are a combination of physical health indicators, cardiovascular risk factors, demographic variables, and engineered composite features. The highest-ranking predictors across models include:

- CardioRisk_Score
- General Health (GenHlth)
- BMI and BMI-Age Interaction
- High Blood Pressure (HighBP)
- BMI Category

- Income and Age Group

As shown in Figure 7, cardiovascular and general health indicators dominate the importance rankings, with engineered features such as CardioRisk_Score and BMI_Age_Interaction consistently appearing among the top predictors.

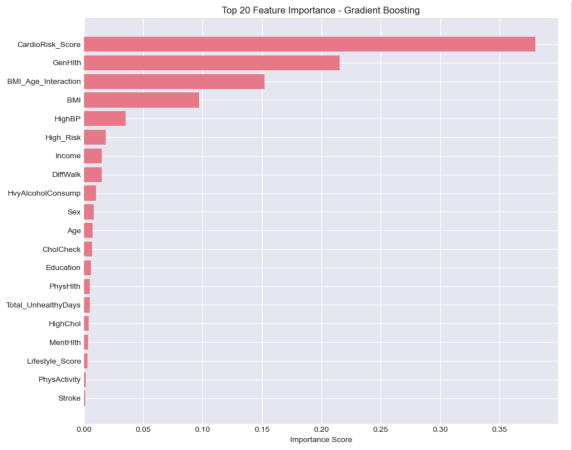

Figure 7 : *Feature importance analysis from XGBoost*

First, physical and metabolic health factors, such as BMI, general health status, and cardiovascular risk, are the strongest indicators of diabetes risk. This aligns with medical understanding that diabetes is closely linked to metabolic and cardiovascular conditions.

Second, demographic and socio-economic factors, including age, income, and education, also play an important role. Older age groups and lower-income populations show higher predicted risk, reflecting the influence of long-term exposure and unequal access to health resources.

Third, behavioral and lifestyle factors, such as physical activity, alcohol consumption, and smoking, contribute to risk and highlight that diabetes is not only a medical condition but also a behavior-related and socially influenced disease.

4.2 Practical Implications

First, diabetes risk is often visible long before a clinical diagnosis is made. Indicators such as cardiovascular risk, general health condition, and weight-related measures already contain meaningful signals. This supports the value of early screening and preventive public health interventions.

Second, diabetes is not only a medical issue but also a social issue. The importance of income, education, and access-related variables indicates that health risks are unevenly distributed across society, and disadvantaged groups may face higher exposure to chronic disease risks.

Third, lifestyle-based and preventive interventions may be more effective and more cost-efficient than treatment alone. Since many risk factors are modifiable, prevention-oriented strategies could substantially reduce the long-term burden of diabetes.

4.3 Implications for Stakeholders

For public health authorities, the model can support low-cost population screening and help prioritize prevention programs for high-risk communities.

For businesses such as insurance companies and digital health platforms, the model can be used to support risk stratification and personalized health recommendations, but should not be used as an automated decision system without human oversight.

For individuals, the findings emphasize that personal health behaviors related to weight, activity, and cardiovascular health can significantly influence long-term diabetes risk.

4.4 Limitations and Ethics

This study has several limitations. The data is self-reported and may contain reporting bias or measurement error. The model identifies correlations rather than causal relationships and therefore cannot establish direct causality.

The model also performs poorly on minority subgroups, particularly the prediabetic class, where recall is close to zero, and only moderately on diabetic individuals. This limits its usefulness for early individual-level risk detection. Future work could address this limitation by incorporating longitudinal health records and by mitigating class imbalance through targeted data collection or resampling techniques.

There are also ethical considerations. Variables such as income and education reflect social inequality, and if used improperly, the model could reinforce existing disadvantages. Therefore, the model should be applied with transparency, fairness, and appropriate human oversight.

4.5 Summary

Overall, this study shows that diabetes risk is strongly associated with a combination of physical, behavioral, and socio-economic factors that are observable and, in many cases, modifiable. The results support a shift from a treatment-centered approach toward a prevention-focused health management strategy. The model should therefore be understood as a complementary decision-support tool rather than a replacement for clinical or professional judgment.

CHAPTER 5 : PLAN FOR REPRODUCIBLE RESEARCH

5.1 Reproducible Research

Reproducible research is an indispensable and crucial part of any research or project. Once reproducibility is implemented in a given research, it further elevates the credibility of that research as other researchers are able to understand, analyse and recreate the outcome to prove the validity. A research can be reproducible if certain elements are present, including the public access to the actual data, the statistical method are fully described, comprehensive documentation and standard method of distribution are employed. On the contrary, if a research is not reproducible, the consequences include loss of trust towards scientific research, a waste of resources as researches will require to spend ample time in attempting to replicate unclear methods. This further underscores the importance of reproducible research.

5.2 Methods to Ensure Reproducible Research

One of the key approaches in ensuring reproducible research is utilising tools equipped with reproducible features which will be detailed in the following parts. Literature programming is a tool that is able to integrate narrative text, code and results in a single document. Modern examples of these include RMarkdown, knitr and Jupyter Notebook which facilitates the seamless integration of documentation and code syntax. Next, a tool that has the capacity to track provenance to identify the origin and chronology of research objects, including data source, source code, figures and results. This ensures that every transformation from raw data to the final result can be traced and verified. Subsequently, a tool that promotes collaboration functionality and infrastructure, allowing accessible, convenient and dynamic real-time teamwork on projects, reducing the friction of passing codes around manually. Moreover, tools that are expressed as computation notebooks are highly considerable since it enables commands and calculations to be executed in an interactive and decoupled manner parts by parts which highly increases efficiency. Besides this, the capability to preserve the processing environment is also an important factor as it ensures that all the analyses and findings remain executable even as software and dependencies evolve.

5.3 Plan for Reproducible Research

The very step in ensuring our project is reproducible is to consciously set that intent from the very start, which we do. Below explains the approaches that were implemented in maintaining the reproducibility of our project.

5.3.1 Selection of Tool with Reproducibility Features

Firstly, we will utilize a tool that satisfies all the five key features outlined above. After extensive consideration, we come to a conclusion that Google Colaboratory (Colab) is the most optimal tool for our project in upholding research reproducibility as it satisfies all the five features:

1. Literate programming: Colab notebooks seamlessly integrate Markdown text, executable Python code, and rich visualizations in a single, coherent document

2. Computation tracking: All code cells maintain execution order and output history, providing clear provenance of computational results

3. Version control: Built-in integration with Google Drive enables automatic versioning, while direct GitHub integration facilitates collaborative version control and public repository hosting.

4. Notebook interface: As a platform based on Jupyter notebook, Colab provides the industry-standard notebook interface for data science workflows.

5. Environment preservation: Google's cloud infrastructure ensures consistent Python environments with pre-installed common data science libraries such as Pandas, Numpy, Matplotlib and Scikit-learn.

5.3.2 Intended Type of Reproducible Research

Aside from selecting the most optimal tool, we intent to integrate the following crucial elements which allow our project to be considered as ‘Open or Reproducible Research’:

1. Well-documented explaining each and every step with clear codes

2. Google Collab notebook that is publicly accessible

3. Use Kaggle dataset that is publicly available

Once the above elements are implemented, it will allow independent researchers to fully audit our outcome, replicate our results and also extend our findings for another research.

5.3.3 Deterministic Computation

Outputs and inferences for machine learning algorithms and models can be classified into two, either deterministic or probabilistic. Deterministic models provide a fixed outcome every single time given a specific input which offers consistency, while probabilistic models incorporate changes, yielding a range of possible outcomes associated with likelihoods. To ascertain that our model produces the same result, we deploy the deterministic model. Specifically, this can be

achieved by setting a specific Random Seed for all of the related processes such as train-test splits.

CHAPTER 6 : DEPLOYMENT OF DATA PRODUCT

6.1 Model Selection and Justification

The proposed data product utilizes the XGBoost Classifier which demonstrated the highest predictive accuracy and F1-score during the experimental phase. This model was selected for its ability to handle the non-linear relationships within the BRFSS 2015 dataset, specifically regarding high-impact indicators such as GenHlth and BMI.

6.2 System Architecture

The deployment is conceptualized as a RESTful API service, facilitating integration with electronic health records (EHR) and mobile health applications.

- Serialization: The trained model and the StandardScaler object are serialized using the joblib library. This ensures that the exact normalization parameters (mean and variance) used during training are applied to live data, preventing feature scaling bias.
- Infrastructure: The model is encapsulated within a containerized environment (e.g., Docker) to ensure environment consistency across development and production servers.

6.3 The Inference Pipeline

To ensure the model remains reliable in a production environment, the deployment pipeline automates the data preparation steps identified in the research:

1. Preprocessing: Raw input features are mapped to the categorical encoding used in the notebook.
2. Transformation: The StandardScaler is applied to continuous variables to align input data with the model's expected distribution.

3. Classification: The model generates a probability score for three classes: Non-diabetic, Pre-diabetic, and Diabetic.

6.4 Performance Monitoring and Model Governance

Post-deployment, the system implements a monitoring framework to maintain clinical validity:

- Metric Tracking: The API logs predictions to monitor for Data Drift, ensuring that changes in the population's health trends do not degrade model performance.
- Explainability (XAI): Utilizing the feature importance rankings derived in the notebook, the product provides a "Reason Code" for each prediction. By highlighting variables like HighBP or Age, the system provides clinicians with transparent, actionable insights rather than a "black-box" result.

6.5 Scalability and Future Maintenance

The architecture supports periodic retraining. As new annual BRFSS data becomes available, the pipeline is designed to re-validate the model against fresh ground-truth labels, allowing for continuous improvement of the diagnostic accuracy without requiring a full system redesign.

6.6 Proof-of-Concept Demo

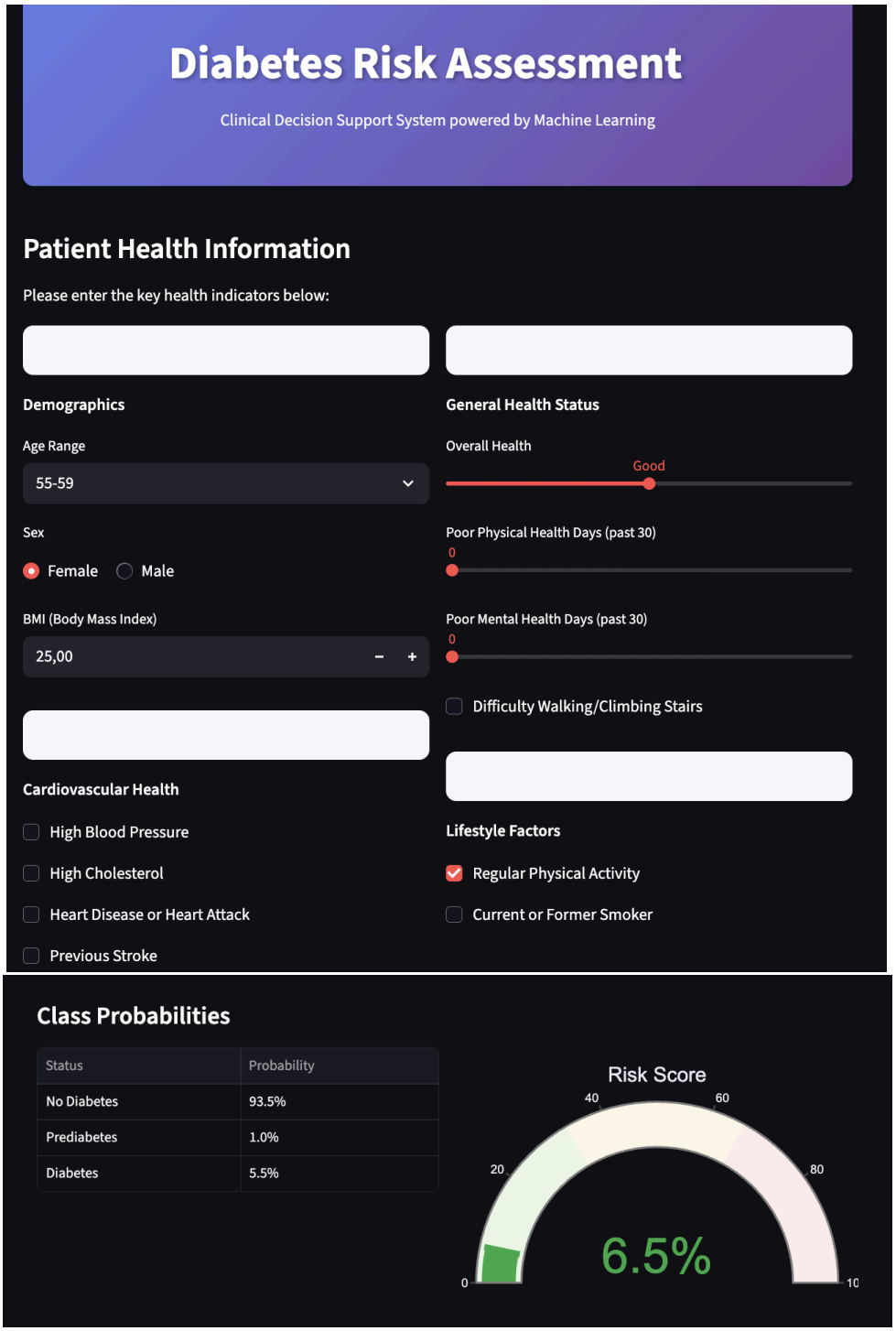

Figure 8 : *Clinical Decision Support System*

This proof-of-concept demonstrates a Clinical Decision Support System built with Streamlit that operationalizes the trained XGBoost model for real-world diabetes risk assessment. Healthcare professionals can input patient health information, including demographics (age, sex, BMI), cardiovascular indicators (high blood pressure, cholesterol, heart disease history), general health status, and lifestyle factors (physical activity, smoking habits)—through an intuitive web interface. The system processes these inputs through the deployed model to generate a comprehensive risk assessment, providing class probabilities for three categories (No Diabetes, Prediabetes, Diabetes) alongside an overall risk score visualized on a gauge chart.

This Streamlit-powered application serves as a clinical reference aid, enabling healthcare providers to quickly identify at-risk patients, prioritize interventions, and facilitate data-driven conversations about diabetes prevention and management. By translating complex machine learning predictions into actionable insights through an accessible web interface, the system bridges the gap between predictive analytics and clinical practice, supporting more informed healthcare decisions without replacing professional medical judgment.

CHAPTER 7 : INSIGHTS AND CONCLUSION

7.1. Key Insights from Data Analysis

The BRFSS 2015 dataset's exploratory analysis identified important trends in the prevalence of diabetes and related health indicators. The substantial relationship between age, body mass index (BMI), and self-reported general health (GenHlth) and the chance of receiving a diabetes diagnosis was a noteworthy discovery. In particular, compared to those without diabetes, those classified as pre-diabetic or diabetic consistently reported worse general health and greater BMI. The research also revealed a distinct socioeconomic gradient, showing that higher rates of the condition were linked to lower levels of education and income, highlighting the significance of taking non-clinical survey data into account when monitoring public health.

7.2. Model Performance and Predictive Power

The comparative analysis of different machine learning algorithms verified that ensemble approaches which are the Random Forest XGBoost are successful for this classification

problem. Despite the dataset's intrinsic class imbalance, these models produced great results across important measures like accuracy, ROC-AUC, and F1-score, demonstrating a strong ability to differentiate between the three target classes. The model is a useful tool for early-stage screening in non-clinical settings because of its high ROC-AUC scores, which in particular demonstrate its dependability in detecting at-risk individuals.

7.3. Conclusion and Future Work

In summary, this project effectively showed that survey-based health indicators can be transformed into a potent predictive diagnostic tool using a whole machine learning pipeline, from data engineering to model validation. The resulting model offers a low-cost, scalable way to identify people who are at risk of developing diabetes, which can help medical professionals better target interventions. In order to improve the accuracy of pre-diabetes identification, future research should concentrate on more aggressively tackling the class imbalance through sophisticated sampling strategies and investigating deeper neural network topologies. The predictive model can significantly lessen the long-term burden of chronic illness by incorporating these findings into actual therapeutic procedures.

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